MINOR PROJECT PRESENTATION

AI-Based Water Resources Management Dashboard

Intelligent Water Management Using AI

- Consumption Analysis and Demand Prediction
- Data-Driven Decision Support System
- Efficient Water Resource Management

Department of Computer Science & Engineering specialition in Artificial inteligence

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PROJECT TEAM

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PROJECT BACKGROUND

Introduction & Problem Statement

Introduction

- Water demand is increasing continuously across regions.
- Water consumption data is growing rapidly every day.
- Manual analysis of large datasets is difficult and slow.
- Different sectors have different consumption patterns.
- AI helps identify hidden consumption patterns effectively.
- Dashboards simplify complex water data into clear visuals.

Problem Statement

- Large water datasets are difficult to analyse manually.
- High-demand zones need better monitoring and tracking.
- Future demand is difficult to predict with static tools.
- Environmental factors need better analysis and correlation.
- Data-driven decision support is limited in current setups.

GOALS & TARGETS

Project Objectives

Core objectives to achieve intelligent water resource management:

- Analyse water consumption patterns effectively across urban sectors.
- Identify high-consumption sectors and critical distribution zones.
- Study environmental factors affecting daily water demand.
- Predict next-day water demand using machine learning regression models.
- Provide explainable analytical insights for municipal authorities.
- Support data-driven management decisions and efficient planning.
- Visualize important findings through an interactive and intuitive dashboard.

PROJECT SCOPE

ANALYTICS
Understand historical trends and sector breakdown

PREDICTION
Forecast next-day municipal demand accurately

DECISION
Empower operators with clear recommendations

COMPARATIVE ANALYSIS

Existing System & Proposed Solution

Existing System

- Traditional systems mainly monitor basic consumption.
- Smart systems provide real-time monitoring without deep insights.
- ML systems enable demand forecasting but run as isolated models.
- Separate tools create information gaps across departments.
- Explainability is often limited for municipal managers.
- Integrated decision support is needed to guide operations.

Proposed Solution

- Combine consumption analysis and ML prediction into one flow.
- Analyse sector and zone consumption with drill-down capability.
- Include environmental factors like temperature and weather.
- Predict future water demand accurately for next-day planning.
- Present insights through one unified interactive dashboard.
- Generate automated recommendations for proactive planning.

DATA OVERVIEW & SCOPE

Dataset & Analysis

Dataset Overview

9,075 **Total Records**

Historical daily consumption observations

12 **Attributes**

Consumption, zones, sectors, weather & time features

5 **Major Sectors**

Residential, Commercial, Industrial, Agriculture, Public

5 **Different Zones**

Geographic municipal water distribution zones

Analysis Highlights

- Compare consumption across sectors to identify major user groups.
- Compare consumption across zones to detect regional variations.
- Analyse consumption trends and periodic patterns over time.
- Identify high-consumption areas requiring infrastructure attention.
- Study relationships between important operational variables.
- Use correlations to understand significant environmental impacts.

WORKFLOW & PROCESS

Project Methodology

STEP 01
Data Collection

→

STEP 02
Data Cleaning & Preprocessing

→

STEP 03
Exploratory Data Analysis

→

STEP 04
Feature Engineering

STEP 05
ML Model Training

→

STEP 06
Demand Prediction

→

STEP 07
Dashboard Visualization

→

STEP 08
Actionable Recommendations

KEY METHODOLOGICAL FOCUS

Historical Data + Environmental Factors ➔ Demand Prediction ➔ Decision Support

Transforms raw municipal consumption records and weather variables into proactive, next-day demand forecasts.

MODEL TRAINING & EVALUATION

ML Model & Results

Machine Learning

- Use regression-based ML models for demand prediction.
- Train models using historical consumption and relevant factors.
- Predict next-day water demand with high reliability.

Results

- Identifies sectoral consumption patterns effectively.
- Detects high-demand sectors and priority distribution zones.
- Accurately predicts future water demand for operations.
- Compares actual and predicted demand to minimize error.

MODEL EVALUATION METRICS & FINAL VALUES

MAE: 0.14 MLD
Mean Absolute Error
Low average deviation across days

RMSE: 0.21 MLD
Root Mean Squared Error
Penalizes large prediction errors

R^2 Score: 0.93
Coefficient of Determination
Strong explanatory and predictive power

Final Model Values: MAE: 0.14 | RMSE: 0.21 | R^2: 0.93 (Baseline: MAE: ___ | RMSE: ___ | R^2: ___)

INTERACTIVE SYSTEM VISUALIZATION

AI Water Management Dashboard

Key Features

- Total water consumption tracking
- Sector-wise consumption breakdown
- Zone-wise consumption comparison
- Historical consumption trends
- High-demand zone identification
- Actual vs. predicted demand tracking
- Environmental factor analysis (Temp)
- Interactive analytics and decision support

Purpose

Convert complex water data into clear and actionable insights.



TECHNOLOGY STACK

Tools & Technologies

1. Programming Language

• Python — Data processing, analysis & Machine Learning

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3. Machine Learning

• Scikit-learn — ML training & evaluation

• XGBoost — Advanced demand prediction

4. Dashboard

• Streamlit — Interactive web dashboard using Python

5. Development Environment

• Jupyter Notebook — Analysis & model development

• VS Code — Project development & coding

6. Dataset

• CSV / Excel — Store and manage 9,075 water-consumption records

PROJECT TECHNOLOGY FLOW

Python → Pandas / NumPy → Scikit-learn / XGBoost → Streamlit

INSIGHTS & PROJECT SUMMARY

Recommendation Module & Conclusion

Recommendation Module

Based on analysis and predictive models:

- Monitor high-demand zones continuously.
- Plan resources before peak demand occurs.
- Investigate unusual consumption patterns or surges.
- Improve municipal resource allocation efficiency.
- Use demand predictions for proactive reservoir planning.

Conclusion

The proposed system combines AI, data analysis, ML prediction, and visualization to understand water consumption and support efficient, data-driven water resource management.

Future Scope

- IoT-based smart meters & real-time telemetry
- Automated pipeline leakage detection
- Live weather API integration for dynamic forecasts
- GIS-based spatial mapping & visualization
- Advanced deep learning forecasting architectures

THANK YOU

AI-Based Water Resources Management Dashboard

"From Water Data to Intelligent Decisions."

Questions & Answers

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